NEUROLAB AI • STEM PHYSICS LABORATORY

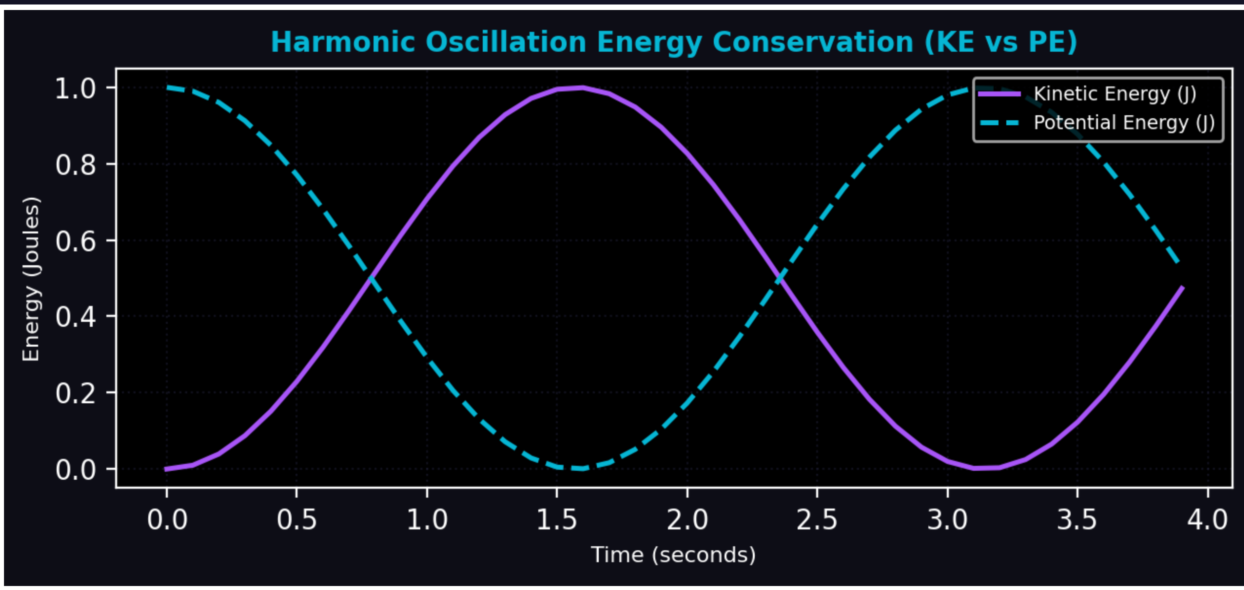
HIGH-RESOLUTION ANALYTICAL CALIBRATION REPORT • GENERATED 2026-05-31 08:45

STUDENT INVESTIGATOR	Marie Curie	**EXPERIMENT CALIBRATION**	PENDULUM HARMONIC SWING
CALIBRATION DATE	2026-05-31 08:45	**METRICS CAPTURE**	LIVE WEBCAM OVERLAY (1080p)

I. RECONSTRUCTED KINEMATIC METRICS

ANALYTICAL PROPERTY	RECORDED VALUE
Average Velocity	3.45
Peak Height	1.25
Measured Frequency	0.63
Calculated Gravity	9.78

II. GRAPHICAL TRAJECTORY VECTOR



III. AI STEM TUTOR GROUNDED ANALYSIS

■■ Pendulum Oscillation Analysis!

I detected a harmonic oscillation! Based on the tracked path, here are the calculated metrics of your pendulum:

For small angles, a pendulum exhibits Simple Harmonic Motion (SHM). The period is independent of the mass and depends only on length (L) and gravity (g).